

```
exercise4.py x
1 class Dancer:
2     def __init__(self, name, nationality, style):
3         self._name = name
4         self._nationality = nationality
5         self._style = style
6
7     def get_name(self):
8         return self._name
9
10    def set_name(self, new_value):
11        self._name = new_value
12
13    def get_nationality(self):
14        return self._nationality
15
16    def set_nationality(self, new_value):
17        self._nationality = new_value
18
19    def get_style(self):
20        return self._style
21
22    def set_style(self, new_value):
23        self._style = new_value
24
25    name = property(get_name, set_name)
26    nationality = property(get_nationality, set_nationality)
27    style = property(get_style, set_style)
```

Encapsulation Exercise 4

Exercise 4

Define the `Dancer` class which has attributes for `name`, `nationality`, and `style`. Use the Python convention to make these attributes private. Create the getters and setters using the `property` function.

Expected Output

Initialize an object of the `Dancer` class as follows:

```
my_dancer = Dancer("Savion Glover", "American", "tap")
```

Task	Output
<code>print(my_dancer.name)</code>	<code>Savion Glover</code>
<code>print(my_dancer.nationality)</code>	<code>American</code>
<code>print(my_dancer.style)</code>	<code>tap</code>
<code>my_dancer.name = 'Anna Pavlova'</code>	N/A
<code>my_dancer.nationality = 'Russian'</code>	N/A
<code>my_dancer.style = 'ballet'</code>	N/A
<code>print(my_dancer.name)</code>	<code>Anna Pavlova</code>
<code>print(my_dancer.nationality)</code>	<code>Russian</code>
<code>print(my_dancer.style)</code>	<code>ballet</code>

Important

Use the property function; follow the Python convention for private attributes.

TRY IT

Command was successfully executed.

Submit your code when ready.

Here is one possible solution:

```
class Dancer:
    def __init__(self, name, nationality, style):
        self._name = name
        self._nationality = nationality
        self._style = style

    def get_name(self):
        return self._name

    def set_name(self, new_value):
        self._name = new_value

    def get_nationality(self):
        return self._nationality

    def set_nationality(self, new_value):
        self._nationality = new_value

    def get_style(self):
        return self._style

    def set_style(self, new_value):
        self._style = new_value

    name = property(get_name, set_name)
    nationality = property(get_nationality, set_nationality)
    style = property(get_style, set_style)
```

Be sure to use a leading underscore with the `_name`, `_nationality`, and `_style` attributes. For each attribute, use the `property` function. The getter should always be the first parameter.

Check It!

LAST RUN on 12/28/2023, 2:06:43 AM

Test passed

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